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Ecoleaders

Empowering teachers for a sustainable future

DIGITAL TOOL KIT

2024-2-EE01-KA210-SCH-000294262





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INTRODUCTION

Creating Eco-Leadership Skills

The digital toolkit developed as a result of the EcoLeaders project aims to support teachers in developing their eco-leadership skills and to equip them with digital tools and technologies that foster sustainability in teaching practices. The goal is to encourage educators to adopt ecological approaches and critical thinking in order to promote environmental awareness in classrooms. This also involves supporting teachers' ecolinguistic skills, which further strengthens their ability to integrate ecological awareness into education and communication more broadly. Our vision is to develop a broader understanding of the role language plays in shaping ecological consciousness, leading to more sensitive and sustainable communication approaches in schools. In addition, the digital toolkit provides teachers with methods for applying game-based learning and digital tools to engage students in green initiatives and sustainability-focused learning.

The EcoLeaders digital toolkit is inspired by the European Commission's Digital Education Action Plan (2021–2027), which provides a common vision for high-quality, inclusive, and accessible digital education in Europe. The Digital Education Action Plan was developed to respond to the educational needs of European Union countries at a time when, according to research, teachers require more training in the educational use of digital tools and many students still lack basic digital skills (OECD, 2018; ICILS, 2023).



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Through the digital toolkit, we aim to encourage teachers to use digital tools and applications effectively in their everyday school practice. Teachers who possess up-to-date digital competences and the ability to apply digital knowledge in teaching are best equipped to teach their students the digital skills required in line with the European Commission's digital objectives.

The EcoLeaders digital toolkit has been developed through international European cooperation, bringing together best practices from schools across different countries. Our vision has been to strengthen partnerships between schools from diverse countries and backgrounds, promoting knowledge exchange and cross-cultural cooperation in green and digital education.



Green education

The integration of sustainability competences and green education into school curricula is one of the key objectives set by the European Union for education across Europe. A more sustainable and greener future for Europe—and indeed for the entire planet—has become an increasingly urgent global issue, requiring solutions to mitigate the accelerating effects of climate change.

The Intergovernmental Panel on Climate Change (IPCC) projects that, if greenhouse gas emissions continue at current levels, the global average temperature is likely to rise by around 1.5°C between 2030 and 2052. This projection has further increased the focus on the role of education in addressing climate change. In line with this, reports from UNESCO highlight the growing importance of climate literacy, emphasizing that understanding climate-related issues is essential for enabling individuals and communities to take informed action on environmental challenges.

The EcoLeaders digital toolkit aims to provide resources for teaching and addressing climate change as part of everyday school practice.



Often, the teaching of sustainability is left to individual schools or teachers, which limits its broader impact. Research indicates that teachers need more resources and additional training to effectively teach sustainability in schools. Around half of young people feel they receive sufficient education on sustainability at school, and although the majority consider it important, fewer than one third actively engage in sustainable actions in their daily lives (European Commission, 2024). Young people therefore need more meaningful sustainability education in order to develop the knowledge, skills, and motivation to act sustainably in their everyday lives. This may also increase their interest in future careers and job opportunities emerging from the green transition.

Through international collaboration, we have aimed to develop resources that support both digital and sustainability-oriented learning. The toolkit presents affordable educational applications and digital tools that teachers can easily integrate into their teaching. It also includes adaptable lesson plans related to sustainability and climate change. The purpose of these resources is to support digital and green learning in schools across different countries in line with the Sustainable Development Goals.

The European Commission's Digital Education Action Plan (2021–2027) highlights the need to strengthen digital competences as part of the green transition. It encourages education systems to make use of environmentally friendly digital solutions that help lower environmental impacts while also supporting climate-related learning.

The EcoLeaders project is closely aligned with these objectives by promoting green digital skills and supporting teachers in using digital tools in ways that enhance sustainability, reduce the use of resources, and improve climate literacy in classrooms. In addition, the wide availability and accessibility of digital technologies is expected to increase the overall reach and effectiveness of the project.

Findings from the European Schoolnet suggest that the integration of digital tools into teaching can improve student learning outcomes, especially in areas connected to global issues such as climate change.



Sources

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GREENCOMP FRAMEFORK

Sustainability and digital transformation

Europe's digital transformation is closely linked to the European Green Deal. In the context of the Digital Decade, the EU is addressing two key priorities at the same time: advancing both the green and digital transitions. Digitalisation enables individuals to reduce their personal carbon footprint in everyday life. Even small daily choices can have a significant long-term impact, as these decisions collectively form a person's overall carbon footprint.

Since every individual can contribute to positive change through their own actions and encourage their community toward, for example, sustainability, learning sustainability competences, environmental values, and attitudes in both formal and informal learning contexts is increasingly important as Europeans move forward with the digital and green transitions.



GreenComp is a framework for sustainability competences that establishes a shared foundation for learners and guides educators by clarifying what sustainability competence involves. It supports education and training across lifelong learning and was developed under the European Green Deal to advance sustainability learning across the European Union. In short, the GreenComp framework defines sustainability as prioritising the needs of all life on Earth and the planet itself, while ensuring that human activity remains within planetary boundaries.

The framework also outlines key competences for education programmes, helping learners build the knowledge, skills, and attitudes needed to think, plan, and act as thoughtful and responsible citizens, in a thoughtful and skilled way that supports the environment and public health.

GreenComp defines a sustainability competence as the ability to internalise sustainability values and understand complex systems, enabling learners to act or advocate in ways that support ecosystem health, promote justice, and contribute to sustainable futures. The definition emphasises building sustainability-related knowledge, skills, and attitudes so learners can consider sustainability in how they think, plan, and act, and live in balance with the planet. It recognises that this competence can be developed through formal, non-formal, and informal learning at all stages of life—from early childhood to adulthood—and is relevant in both personal and collective contexts.



At the core of GreenComp are 12 sustainability competences that form the foundation for sustainability skills, divided into four main areas. The first area focuses on embracing sustainability values, including valuing sustainability, supporting fairness, and promoting nature. The second area addresses understanding the complexity of sustainability, covering systems thinking, critical thinking, and problem framing.

The third area concerns envisioning sustainable futures, such as futures literacy, adaptability, and exploratory thinking. The fourth area highlights acting for sustainability, including political agency, collective action, and individual initiative. (European Commission, GreenComp, 2022)

The Ecoleaders project's digital toolkit has been developed in line with the GreenComp framework's definition of sustainability and its areas of sustainability competences. In our digital toolkit, we have integrated the GreenComp recommendations for teaching and learning about sustainable development. We have taken into account the teaching of sustainability competences defined by GreenComp as part of the resources in the digital toolkit. We particularly highlight values such as the appreciation of sustainability and fairness, as well as critical thinking and understanding the complexity of sustainability.

Sources:

European Commission. (2022). GreenComp: The European sustainability competence framework. Publications Office of the European Union.

European Commission. (n.d.). Green and digital transition. Digital Strategy. <https://digital-strategy.ec.europa.eu/en/policies/green-digital>



CANVA

Canva makes it possible to create visual presentations online, and it is particularly well suited for producing presentations related to environmental topics. It is a free online graphic design tool. The paid version provides access to additional features, but the free version is also highly suitable for creating visual materials. With Canva, users can design presentations, posters, social media posts, videos, infographics, and much more. Canva's ready-made templates, images, and graphic elements make it easier to create visually appealing presentations.

Both teachers and students can use Canva's online tools to create presentations. Canva is user-friendly, and presentation templates can easily be downloaded and saved in different formats.



Canva for Education is particularly suitable for group work because several students can work on the same project simultaneously, and all changes are visible immediately. Students can create presentations, infographics, written texts, videos, and posters related to environmental themes. Among the Canva for Education tools, the Whiteboard feature is especially useful for collaborative brainstorming before starting group work. Mind maps, sticky notes, images, arrows, drawing tools, and writing functions enhance collaborative brainstorming activities. The Docs tool is a great tool for writing essays, notes, and presentations. Images and tables can be added to the text, and the text can be transformed into a slideshow presentation with a single click using the Slides feature.



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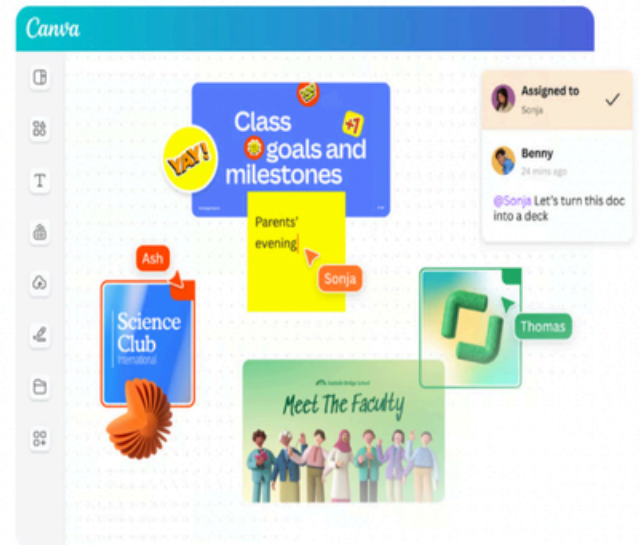
Education Overview Teachers Students **Schools** Higher education Case studies

Canva for Schools

Empower teachers, students and staff to collaborate and communicate visually with Canva Education, 100% free for primary and secondary schools. Unlock Canva Education for your school or trust today.

Get started

Watch webinar



Students can record their own presentations using the Present & Record feature. The tool records audio and, if desired, video as well. The recording can be saved as a link that teachers and classmates can view later. Canva for Education's Magic Write and AI tools help generate and develop text ideas. Canva's AI features can be used to ask for ideas and explanations, summarise text, and improve text flow. These tools provide useful support for writing, but students should still be encouraged to review and edit the text themselves. Canva for Education includes all premium features free of charge. It is also free for students when the teacher shares an invitation link, for example via WhatsApp or email.



Creating Canva for Education Access for a School

To create access for your school to Canva for Education, go to canva.com/education and select “Get verified” to apply for Education access. Log in using your school email address, fill in the required information about your school, and complete the verification process. After these steps, you can wait for your information to be approved. Once Canva has verified the information, you will gain access to the Education account. After activating the Education account, teachers can select the “Classroom” section in Canva and create and name a new class, for example an environmental project. Students can be invited to the project through an email invitation, invitation link, or Google/Microsoft classroom code. Students join the class by clicking the invitation link. Within the classroom, teachers can share ready-made templates, create assignments, and set deadlines for submitting work.



Learning activity



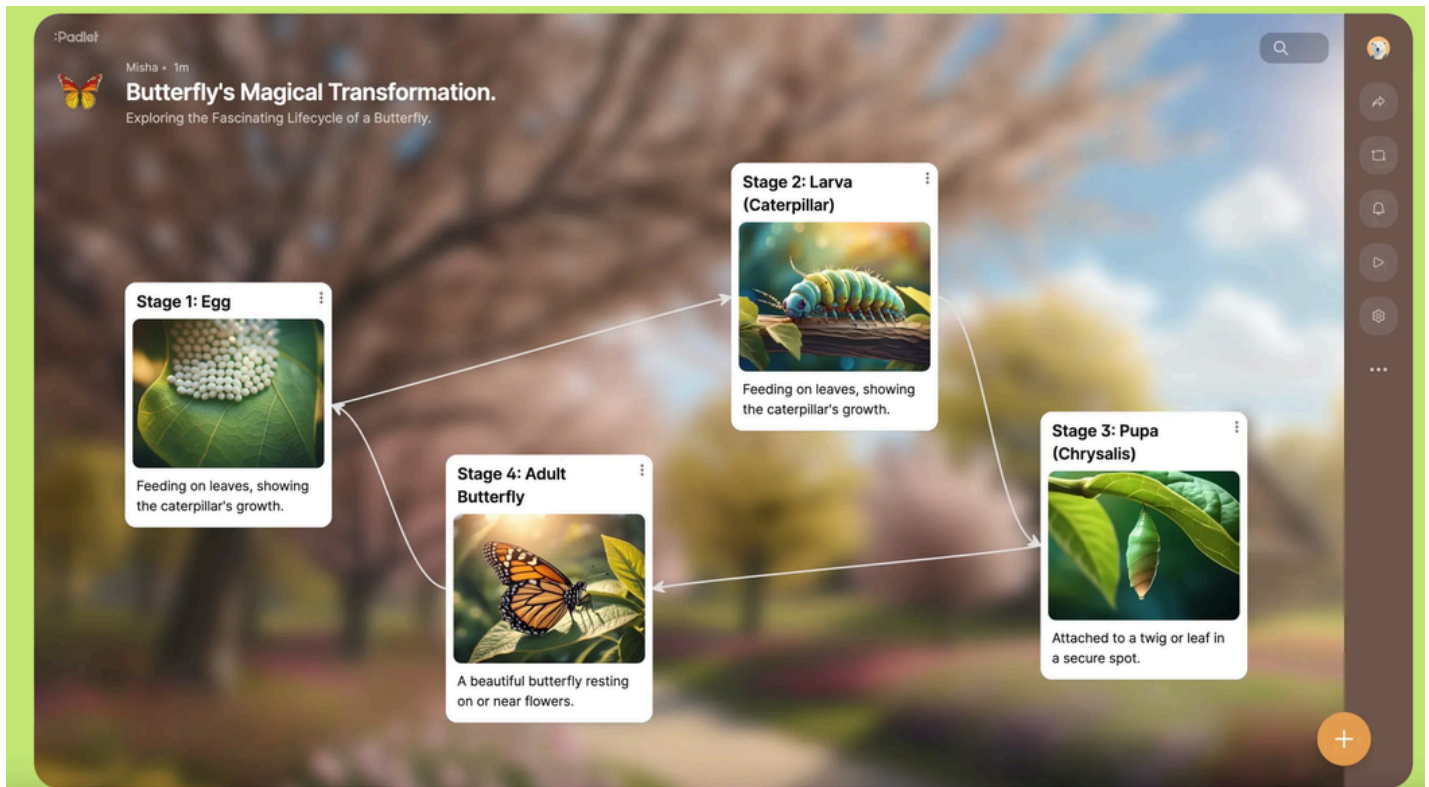
The teacher can ask students to create a group PowerPoint presentation on current environmental topics, such as:

- The importance of recycling
- Tips for saving energy
- Reducing the school's carbon footprint
- The consequences of climate change
- What Earth Hour is and why it is celebrated
- The importance of local food
- The harms and consequences of air pollution

Students should be encouraged to make use of Canva for Education's different features when creating their presentations and to include images and infographics to support their work. The presentation should have a clearly defined length, for example 6–8 slides.



PADLET



Padlet is an online, digital, interactive wall where teachers and students can add content and collaborate on ideas together. Padlet is a modern digital version of a notice board. It is easy to use, and using it for learning does not require advanced digital skills. The free version of Padlet has a limited number of boards, which is important to consider when planning teaching activities.

Using Padlet to support teaching is simple. The teacher creates a wall, or "padlet," and shares a link or QR code with students. It is an ideal platform for sharing assignments and other educational content.

Padlet can also be used to support distance learning. Students can participate in creating and editing the Padlet using their own devices. It works directly in a browser without installations and is also compatible with mobile devices.

The wall can be used to add text, images, videos, links, audio recordings, files, and drawings. It is a practical platform for collaborative brainstorming, as all participants can create and collect shared content in one place. Padlet is therefore an ideal tool for group work and portfolio-style projects.



How to Get Started with Padlet

Getting started with Padlet is easy. A user account is created at <https://padlet.com> by selecting “Sign up” in the top right corner of the page. Users can register using an email address, Google account, or Microsoft account.

After logging in, the teacher can create the first Padlet by selecting “Make a Padlet.” Different structures can be chosen depending on the intended purpose. The “Wall” layout is a free-form notice board suitable for collaborative brainstorming. In the “Shelf” layout, content is divided into columns, which works well for group work and step-by-step tasks. The “Timeline” layout allows content to be organised chronologically, while the “Canvas” layout allows content to be connected with arrows and networks in a mind map style.

Next, the teacher names the Padlet and writes a short task description. Clear instructions help students understand the task correctly. The teacher can also adjust students’ permissions in the settings.



If students are assigned the role of readers, they can view the educational content but cannot write or edit it themselves. If they are assigned the role of writers, they can post text and add images and other content, but they usually cannot edit other students' posts. As moderators, students can approve posts, edit content, and manage discussions. The moderator role is suitable, for example, for a group leader. The admin role is usually managed only by the teacher, as it provides permission to edit the wall, delete content, and change settings.

If the teacher wants to review posts before they are published, content moderation can be enabled in the settings. Teachers can also allow likes, stars, voting, and ratings, which are useful for group work, collaborative brainstorming, and different kinds of voting activities.

Padlets can be shared with students via a link, QR code, Google Classroom, or Microsoft Teams. Students can often access the platform without creating separate accounts. Students add content using the plus button by writing text and adding images, videos, and links. The teacher can monitor students' progress in real time from their own device.

Learning activity

Students use their own phones, tablets, or computers during the activity. The teacher creates a Padlet using the Shelf layout to support group work. The teacher creates separate columns for the following ideas:

- Eco-art
- Recycling campaign
- Ecological volunteering day at school
- School flea market / charity sale



At the beginning of the lesson, the teacher shows images of eco-art, famous recycling campaigns carried out for the benefit of the environment, examples of environmental clean-up activities, or ideas for flea markets and second-hand sales. The teacher then asks students whether they have ever done something to help the environment—for example, created eco-art, participated in an eco-art exhibition or recycling campaign, taken part in volunteering activities, or visited a flea market. Students write their answers on the Padlet. Next, the teacher divides the students into groups and assigns each group one idea:

1. **Eco-art**
2. **Recycling campaign**
3. **Ecological volunteering day at school**
4. **School flea market / charity sale**

Each group is asked to come up with five ideas or concrete actions related to their topic. For example, how would they organise an eco-art exhibition or a recycling campaign? What kinds of activities would take place during an ecological volunteering day at school? How would a school flea market be organised? Students may support their ideas by adding images, links, and videos alongside their written text. At the end of the activity, groups comment on and vote for the ideas created by the other groups. They discuss what is good about the ideas and what could still be improved. Finally, the class holds a joint discussion about the topics and which ideas could realistically be implemented at school.



GOOGLE EARTH, ENVIRONMENTAL EXPLORATION



Google Earth is a free online tool for teaching a variety of environmental topics. It allows students to explore the Earth's geography, biology, and cultures through 3D images, satellite imagery, and historical images. Google Earth is particularly suitable for teaching climate change because it enables users to observe environmental changes. For example, it can illustrate glacier melting and urban growth. In educational use, it is also possible to create projects and measure distances.

Google Earth supports visual, location-based, and interdisciplinary learning. In teaching, it develops students' inquiry-based and phenomenon-based learning skills, which promote media literacy and critical thinking. Students explore different locations and regions through visual content while making observations and comparisons. Google Earth also includes ready-made educational materials and tools to support teaching. The Voyager section offers ready-made educational content and virtual field trips covering topics such as nature conservation and climate change.



The Timelapse feature visualises changes on Earth over time, allowing students to examine phenomena such as deforestation or industrialisation. Using the measurement tools, students can measure distances between regions and countries, estimate surface areas, and compare different areas with one another.



Creating Stories in Earth

Google Earth's creation tools allow you to create your own projects. Add points, text, and rich multimedia content to share stories and maps visualized on a 3D globe.

[GETTING STARTED GUIDE](#) →

[SEE STORIES CREATED FOR CLASSROOMS](#) →



Getting Started with Google Earth for Teachers

Teachers can use the browser-based version of Google Earth. Using the tool does not always require signing in with a Google account. The Chrome and Edge browsers provide the best support for Google Earth. Teachers can begin exploring the tool by using the search function to find cities, countries, rivers, or even the location of the school. In the Voyager section, teachers can explore ready-made educational materials related to cities, cultural landmarks, nature conservation, and climate change.



Explorers: Great Waters

Follow six bold navigators as they open the globe to exploration and claim territory for European empires.

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Explorers: Vikings

Five hundred years before Christopher Columbus crossed the Atlantic, Eric the Red and his son Leif Ericson founded settlements in Greenland and North America.

[DISCOVER WITH →](#)

[LESSON PLAN →](#)



Extreme Biodiversity on the High Seas

Join Mission Blue on a high seas adventure to the Costa Rica Thermal Dome Hope Spot where we follow marine biologists in the field tagging turtles, tracking sharks and more.

[HOP ABOARD →](#)



Getting Started with Google Earth for Teachers

In the Projects section, teachers can create and edit their own educational content. Teachers can write instructions, add images and videos, place markers on maps, and create virtual field trips. The Map Style settings help users understand area boundaries, place names, roads, and terrain features.

Using Google Earth largely depends on navigating within the platform, so teachers should familiarise themselves with how to teach students to zoom, rotate the map, and explore 3D and Street View modes. The Timelapse feature also makes it easy to demonstrate changes over time, such as coastline changes or urbanisation.

Learning activity

Investigating Climate Change with Google Earth

Required tools: computer/tablet, Google Earth (and optionally Padlet for compiling findings)

The teacher first shows students a Google Earth Timelapse example, such as glacier melting or deforestation. Students comment on the changes they observe and discuss what may have caused them.

Next, the teacher divides students into groups and assigns each group a topic, such as deforestation, glaciers, or urbanisation. Students must locate examples of their assigned phenomenon in Google Earth and identify visible changes. They discuss the causes and consequences of the phenomenon and record their observations.

Finally, each group presents its topic, the observations made, and the consequences the phenomenon may have in relation to climate change.



AI TOOLS, SUSTAINABLE FUTURE SCENARIOS

AI tools (Artificial Intelligence tools) can be used as part of teaching sustainability-related topics. Since schools aim to develop students' sustainability competences so that they can act responsibly for the environment as they grow into adulthood, it is important to teach them about sustainable future scenarios and how to evaluate them realistically.

Sustainable future scenarios help students understand the environmental impacts of different ways of acting. Positive outcomes resulting from changes in behaviour encourage students to adopt a positive attitude toward actions aimed at protecting the environment and mitigating climate change.

For example, if students construct a future scenario in which consumption continues at its current level, they can calculate how many Earths would actually be needed to sustain human consumption in relation to the planet's carrying capacity. What kind of future can be expected if consumption continues at the same level? What would happen if transport emissions decreased rapidly? How would this affect, for example, air quality or the cleanliness of the oceans?

Students can also evaluate what life in cities might be like in the year 2100, and whether changes are expected in energy sources, food production, or modes of transport in society. What changes are likely to occur, and which of them would be positive? What role does digitalisation play in these changes?

With the help of AI, students can imagine, for example, what Europe might look like in 2100 if it were fully carbon neutral, or what would happen if global emissions were reduced enough to prevent glacier melting. Is it possible to prevent rising temperatures and forest fires in Southern Europe by reducing emissions?



Students can use AI to create two alternative scenarios: one representing a sustainable future and the other an unsustainable future. They can compare the effects of these scenarios in terms of emissions, quality of life, the state of nature, and the role of technology in society.

AI tools help model and visualise future scenarios. Images and infographics of future scenarios help students understand the environmental impacts of different actions, including climate impacts. Students develop critical thinking skills and learn to reflect more broadly on the importance of legislation and political actions in promoting sustainability and protecting the environment.

Using AI Tools to Support Future Scenarios

There are various AI tools that can be used in planning future scenarios. These can be divided into text-based tools, image-generation tools, map-based spatial modification tools, data-based climate simulations, and AI-powered platforms used for presenting results. In this section, we focus on text-based AI, which is the most commonly used form of AI in education.

Text-Based AI Tools

Text-based AI tools help students brainstorm and explain future scenarios in written form. They also provide information to support scenarios and help students gather knowledge about the topic being studied. ChatGPT (OpenAI) is a popular text-based AI tool that is easy for students to use.



ChatGPT Edu



ChatGPT Edu is a version of ChatGPT designed for educational use, taking into account safer and more privacy-conscious use of artificial intelligence. It offers controlled administration, authentication, and user management features.

If a teacher wants to introduce ChatGPT Edu in a school, the process can begin by visiting openai.com and selecting “Get ChatGPT Edu” or “Contact Sales.” The teacher then fills in the school’s details using the form provided by OpenAI.

During implementation, the educational institution appoints one or more main administrators for ChatGPT Edu. The school may have designated IT staff members for this role, or responsible persons can be assigned separately within the school. The administrator’s task is to deploy the service using the OpenAI administration panel, through which they can manage settings, users, and service usage, and act as the contact person with OpenAI.



The institution's email domain is verified for user authentication. After verification, users can sign in using their institutional accounts (for example Microsoft 365 or Google accounts) via Single Sign-On (SSO). Necessary user accounts are then created for teachers and students, and appropriate access rights are assigned.

Before using AI in lessons, students should be reminded that AI is a useful tool for searching for new information, but its outputs should not be copied directly without personal reflection. Text-based AI can be used for information retrieval, comparing information, editing texts, and generating ideas, but it is not error-free. Its responses must be evaluated critically and corrected when necessary.



LESSON PLAN ECOLOGY AND ROBOTICS

These lesson plans connect ecological education with educational robotics, helping students understand environmental challenges through hands-on design, programming, observation, and problem-solving activities.

Lesson Plan: Exploring Ecosystems Through Robotics

Grade level: Ages 12–15

Duration: 40' + 40' + 40' + 40'

Theme: Ecology and Robotics

Materials: Robotics kit (e.g. LEGO Spike, mBot, or similar), tablet or computer, worksheets, cardboard, markers, recycled materials, projector

Learning Objectives

- Define the concept of an ecosystem and explain the relationships between living organisms and their environment.
- Identify ecological problems such as pollution, habitat loss, and biodiversity decline.
- Use robotics to model ecological interactions and environmental challenges.
- Develop problem-solving and teamwork skills through the design and programming of a robot-based ecological solution.



Lesson Development

1. Engage

The teacher begins by asking the class: "How can robots help us understand and protect nature?" Students share examples such as robots used to monitor forests, clean oceans, or collect environmental data. The class watches a short introductory presentation about ecosystems and environmental challenges. Students then brainstorm living and non-living elements of an ecosystem and discuss how these elements are interconnected. To activate prior knowledge, students create a simple concept map linking plants, animals, water, soil, climate, and human actions.

2. Explore

Students are divided into small groups and introduced to the challenge: create a robotic model that represents an ecological issue or process. Possible ideas include a robot that simulates an animal moving through its habitat, a robot that detects pollution in a model environment, a robot that helps sort waste for recycling, or a robot that represents the movement of pollinators between flowers. Each group discusses its idea, sketches a simple design, and identifies the ecological concept behind the model.

3. Explain

The teacher presents key ecological concepts such as food chains, habitats, biodiversity, pollution, sustainability, and conservation. At the same time, the teacher introduces the robotics concepts needed for the task, including motors, sensors, simple programming sequences, and iterative testing. Students explain how their robot can represent or respond to a real environmental problem and write a short summary of their idea.



5. Evaluate

Students are assessed through their participation in discussions and group work, the quality and functionality of the robotic model, the accuracy of the ecological explanation, and the creativity of the proposed solution. As a final reflection, students answer the question: "How can robotics help protect ecosystems in real life?" This reflection may be presented in person, on a poster, or in a short digital slide.



LESSON PLAN – ENVIRONMENTAL MONITORING WITH ROBOTICS

Lesson Development

Grade level: Ages 12–15

Duration: 40' + 40' + 40' + 40'

Theme: Ecology, Data Collection and Robotics

Materials: Robotics kit with sensors, tablets or computers, observation sheets, access to the schoolyard or garden, recycled materials, measuring tape

Learning Objectives

- Recognize the importance of observing and monitoring the environment.
- Identify how technology and robotics can support ecological research.
- Collect and interpret basic environmental data.
- Design and program a robot or sensor-based system to support environmental monitoring.

1. Engage

The teacher starts by asking: "How do scientists know if an environment is healthy?" Students suggest answers related to observing plants, animals, temperature, water quality, or waste levels. The class is introduced to real examples of environmental technologies such as ocean robots, forest-monitoring drones, or sensors used to monitor air quality and soil moisture. Students discuss how robotics can help humans collect ecological information more accurately and efficiently.



2. Explore

Students go to the schoolyard, garden, or another nearby outdoor area and observe the environment. They look for signs of biodiversity, human impact, waste, plant health, and possible ecological problems. In groups, students record simple observations such as the number of plants or insects seen, the presence of litter, sunny or shaded areas, dry or wet soil, and signs of human disturbance. Back in class, students discuss what kind of environmental data a robot could help collect in that space.

3. Explain

The teacher explains the purpose of environmental monitoring and introduces simple robotics tools such as distance sensors, light sensors, colour sensors, and programmable movement. Students learn that robots can support ecology by collecting data in hard-to-reach places, monitoring changes over time, reducing human error, and supporting faster responses to environmental threats.

Each group chooses one monitoring function for its robotic project, such as detecting waste on the ground, measuring light in different parts of the schoolyard, or moving through a model area to identify habitat conditions.



4. Elaborate

Students design, build, and program their robots. They test them in a simple model environment or, where possible, in the real observation space. They focus on making the robot complete a clear task, linking that task to a real ecological need, improving the model after testing, and recording the results. Each group prepares a short presentation describing the ecological problem or need, the robot's function, the data it collects or the task it performs, and how it could help protect the environment.

5. Evaluate

Students present their projects to the class. Assessment may include understanding of the ecological issue, effective use of robotics and programming, the quality of teamwork and design, and the ability to explain how technology supports ecology. For the final task, students complete a short reflection titled: "Robots for Nature: How can technology help us care for the environment?"

Teaching note: These plans can be adapted to different robotics platforms and can be simplified or extended depending on students' age, prior knowledge, and available equipment.



LESSON PLAN – BIODIVERSITY: UNDERSTANDING THE RICHNESS OF LIFE

Lesson Development

Grade level: Ages 12–15

Duration: 40'+40'+40'+40'

Materials: Natural element cards, tablet, bingo card, online links, pen, paper, string

Preparation: The teacher creates accounts on Edpuzzle, Canva, Mentimeter, and Padlet. Prepares for their use in class.

Learning objectives

- Defines the concept of biodiversity.
- Provides examples of the importance of biodiversity.
- Identifies the main reasons threatening biodiversity.
- Proposes solutions for the sustainability of biodiversity.

1.Engage

Write the name of each natural element listed below on a card (one for each student)..

- Bumblebee
- mushroom
- spring
- grass
- mouse
- worm
- ant
- spider
- river
- sun

- butterfly
- flower
- tree
- snake
- leaf
- owl
- squirrel
- soil
- deer
- woodpecker
- rain
- dead leaf



Form a circle and place the cards in the middle of the circle. Each student picks a card and holds it up in the air so everyone can see the name of the natural element. The student who picks the tree card is given a ball of string. The student holds the ball of string by the end and throws it to someone in the circle. The student who catches the ball explains how the natural element on their card interacts with the tree. Then, they pass the string over their finger and throw the ball to the third student. The student who catches the ball explains how the natural element interacts with the second student's natural element. This continues until everyone has caught the string. The string forms a complex web connecting everyone in the circle. A natural element is selected, and the group discusses what would happen if it were removed from the web and which natural elements would be affected. The group discusses what effect cutting the string with scissors would have on the web that holds everyone together. The complex combination of various elements such as plants, animals, microorganisms, the atmosphere, soil, and water, and the formation of natural habitats is defined as an ecosystem. Students take the test found at the link below in pairs. Students' prior knowledge of biodiversity is determined.

<https://www.amnh.org/explore/ology/biodiversity/what-do-you-know-about-biodiversity2>

2. Explore

An interactive video is watched. A brainstorming session on general concepts related to biodiversity is held, and the questions in the video are answered with student participation.

<https://edpuzzle.com/edpuzzle-originals/63a45b464266ed3cfaf12dab?language=en>

Students create a word cloud by writing three words that describe biodiversity on Mentimeter. The five most frequently written words are identified.



3. Explain

An interactive video featuring biodiversity practices is watched. Students assess species diversity in ecosystems and answer questions about the environmental benefits of biodiversity.

<https://edpuzzle.com/edpuzzle-originals/65c3dae412725595b46189d3?language=en>

If students need to conduct research or work on the concept of biodiversity, they are divided into groups. The groups gather more information about biodiversity by examining the digital resources provided below. Only one or a few of these resources may be provided.

<https://www.amnh.org/explore/ology/biodiversity/what-is-biodiversity>

<https://www.youtube.com/watch?v=ixf33dQDIXc&t=253s>

https://www.youtube.com/watch?v=GK_vRtHJZu4

<https://www.youtube.com/watch?v=y18o0mACCQs>

4. Elaborate

Students are divided into 5 groups. Each group is assigned one of the animals featured in the biodiversity games (red squirrel, rainbird, spiny-backed ray, frog, mullet). The groups follow the instructions and play the games, gathering information about the species and the threats they face. They determine the measures that can be taken against these threats.

<https://biodiversitygames.biologia.unipd.it/>

Each group provides information about the work they have done under the leadership of one student.



5. Evaluate

Students choose one of the topics—5 reasons threatening biodiversity or 5 measures to protect biodiversity—and create an infographic on Canva. During this process, they can conduct research using the information they gathered in the previous activity and the links provided below.

<https://www.amnh.org/explore/ology/biodiversity/going-going-gone>

<https://www.amnh.org/explore/ology/biodiversity/going-going-gone/what-causes-extinction>

<https://www.baamboozle.com/game/2019197>

<https://www.50waystohelp.com/>

<https://www.youtube.com/watch?v=2RC3Hsk90t8&t=1s>

<https://www.calacademy.org/search?q=biodiversity>

The prepared infographics are displayed on a digital board (padlet).

Students take the test found at the link below in pairs. Students' knowledge of biodiversity is assessed, and questions are answered.

<https://www.footprints-science.co.uk/index.php?quiz=Biodiversity>

Students are given bingo cards (Appendix 1). The activity can be conducted at school if the schoolyard and surrounding area are suitable. If not, students are directed to fill out their bingo cards at home, in a park, garden, etc. They are ensured to be prepared for the next lesson.



References/Links

<https://www.amnh.org/explore/ology/biodiversity/what-do-you-know-about-biodiversity2>

<https://edpuzzle.com/edpuzzle-originals/63a45b464266ed3cfaf12dab?language=en>

<https://www.mentimeter.com>

<https://edpuzzle.com/edpuzzle-originals/65c3dae412725595b46189d3?language=en>

<https://www.amnh.org/explore/ology/biodiversity/what-is-biodiversity>

<https://www.youtube.com/watch?v=ixf33dQDIXc&t=253s>

https://www.youtube.com/watch?v=GK_vRtHJZu4

<https://www.youtube.com/watch?v=y18o0mACCQs>

<https://www.canva.com/>

<https://www.amnh.org/explore/ology/biodiversity/going-going-gone>

<https://www.amnh.org/explore/ology/biodiversity/going-going-gone/what-causes-extinction>

<https://www.baamboozle.com/game/2019197>

<https://www.50waystohelp.com/>

<https://www.youtube.com/watch?v=2RC3Hsk90t8&t=1s>

<https://www.calacademy.org/search?gq=biodiversity>

<https://www.footprints-science.co.uk/index.php?quiz=Biodiversity>



BIODIVERSITY BINGO

5 or more ants 	A bird HOPPING on the ground 	3 YELLOW flowers	An animal with 4 legs	A leaf with WAVY edges 
A leaf BIGGER than your hand	A WHITE flower	An animal MAKING a sound 	A leaf with SMOOTH edges 	An animal with 6 legs
An animal with 2 legs	A FLYING animal 	A flower with 6 or more PETALS 	An animal with 8 legs 	A leaf with POINTY edges 
An animal EATING a plant	A tree with SMOOTH bark 	A flower with 2 or more COLORS	A worm 	A tree with ROUGH or BUMPY bark
A tree TALLER than you are	A moth or butterfly 	Grass 	A leaf SMALLER than your hand	A leaf divided into SMALLER leaves



LESSON PLAN: IN PURSUIT OF BIODIVERSITY: A JOURNEY THROUGH NATURE AND DATA

Lesson Development

Grade Level: Ages 12–15

Duration: 40'+40'+40'+40'

Materials: Tablet/smartphone, pen, eraser, paper, binoculars, marker, additional observation area 1, additional observation card 2

Preparation: Accounts are created for applications such as iNaturalist, WWF forests and free rivers, picturebird, ebird, plantnet, and pictureinsect. The applications are downloaded onto devices and prepared for use in class.

Objectives

- Provides examples of biodiversity from different ecosystems.
- Explains the positive and negative effects of human activities on biodiversity.
- Describes the role of technological applications in protecting biodiversity.
- Records biodiversity data in their surroundings using digital tools.
- Collects information from global biodiversity databases.

1. Engage

The activity can be carried out in two ways: by projecting the device on which the application is used, or by having students use the devices in groups.

The WWF Forests augmented reality (AR) application connects you to the natural world with impressive details, allowing you to observe the impact of human actions on forests, no matter how distant they may seem.

<https://www.worldwildlife.org/resources/activities/educational-apps/wwf-forests/>

Using the WWF Free Rivers augmented reality (AR) app, you can observe how wildlife, people, and flowing rivers affect each other; you realize that a sustainable future is only possible if this interaction continues in a healthy way.

<https://www.worldwildlife.org/resources/activities/educational-apps/wwf-free-rivers/>



2. Explore

The following resources are shared so that students can gather information about different species and learn about international organizations related to biodiversity.

The International Union for Conservation of Nature's (IUCN) Red List of Threatened Species is the world's most comprehensive source of information on the global conservation of animal, fungus, and plant species.

<https://www.iucnredlist.org/>

The Global Biodiversity Information Facility (GBIF) is an international network and data infrastructure funded by governments worldwide, aiming to provide open access to data on all life forms on Earth for everyone, everywhere.

<https://www.gbif.org/>

With the Beaty Biodiversity Museum Data Portal, 754,321 specimens from 6 collections showcasing biodiversity from around the world are available online.

<https://collections.beatymuseum.ubc.ca/>

It is noted that many of these resources were shaped and developed collectively with the contributions of scientists and nature observers living in different regions.



3. Explain

To prepare students for biodiversity observations and enable them to gain experience, they are provided with the opportunity to observe certain bird species in a digital environment. Students discover 16 bird species by recognizing their characteristics and listening to their calls. They mark the numbers of the birds they observe on the observation area provided in Appendix 1.

<https://www.amnh.org/explore/ology/zoology/birdwatching-with-theodore-roosevelt>

4. Elaborate

A visit is made to a nearby lake, wetland, or park with the students. The map of the visited area is reviewed with the students before the visit.



Two observation areas are designated for students to make observations related to biodiversity. Students are divided into two groups and take turns observing in these areas:

1. *Observation Area: A 20-square-meter observation area is prepared for bird watching. In this area, bird biodiversity is observed with the naked eye and binoculars. The species of birds observed are identified using the Picturebird and eBird applications, information about their characteristics is collected, and the observation cards in Appendix 2 are filled out. Students draw a sketch of the area they are in, assign a number to each species they observe, and record this information on their own maps.* <https://picturebirdai.com/> <https://ebird.org/region/TR-09/bird-list>



Observation Area: A 30-square-meter area is marked off with colored markers. The biodiversity of plants and insects is observed in this area. The species of plants and insects observed are identified using the Plantnet and pictureinsect applications, information about their characteristics is collected, and the observation cards in Appendix 2 are filled out. Students draw a sketch of the area they are in, assign a number to each species they observe, and record this information on their own maps.

<https://identify.plantnet.org/tr>

<https://pictureinsect.com/>

Observation cards are later recreated digitally on canvas along with the photos taken and shared on the Padlet page.

5. Evaluation

Students are provided with information about the iNaturalist app. Students enter data about the plant, insect, and bird species they have observed into the app.

<https://www.inaturalist.org/>

References/Links

<https://www.worldwildlife.org/resources/activities/educational-apps/wwf-free-rivers/>

<https://www.worldwildlife.org/resources/activities/educational-apps/wwf-forests/>

<https://www.youtube.com/watch?v=TqBEpw8HfZY>

<https://www.iucnredlist.org/>

<https://www.gbif.org/>

<https://collections.beatymuseum.ubc.ca/>

<https://www.inaturalist.org/>

<https://identify.plantnet.org/tr>

<https://pictureinsect.com/>

<https://picturebirdai.com/>



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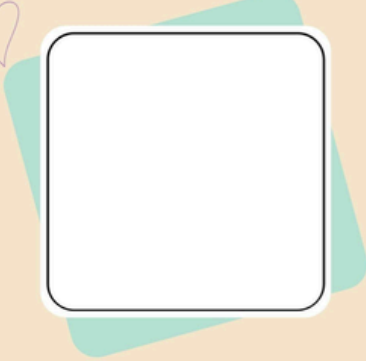
Ecoleaders
Empowering teachers for a sustainable future

Annex 1





Annex 2

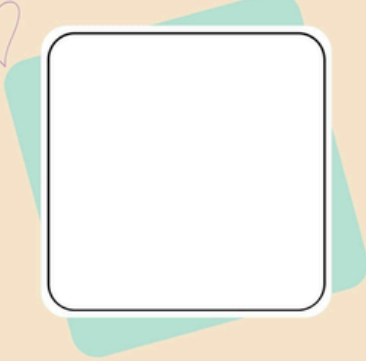


Type Name:

Observation Location:

Observation Time:

Feature:

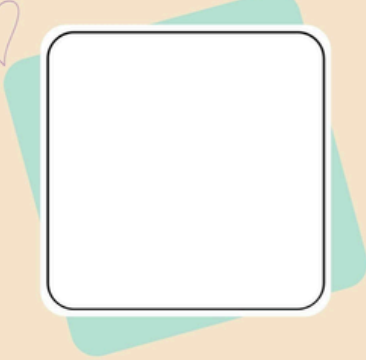


Type Name:

Observation Location:

Observation Time:

Feature:

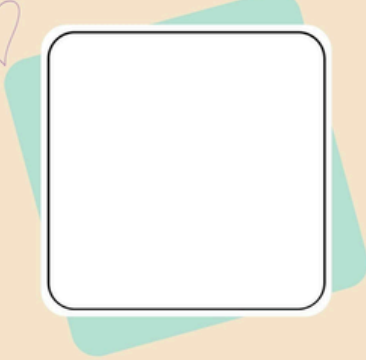


Type Name:

Observation Location:

Observation Time:

Feature:



Type Name:

Observation Location:

Observation Time:

Feature:



Evaluation form

Can YOU help biodiversity ? How many of these jobs can you take on to help biodiversity?

	I could do this	I am doing this
Make wildlife welcome. Support the birds, reptiles, mammals, and plants that live in your neighborhood. You can also attract more wild species by providing water, food, shelter, and privacy.	☐	☐
Protect Habitats. Explore <u>habitats</u> in your area. Help clean up and protect beaches, parks, reserves, and fields where wild plants and animals live.	☐	☐
Volunteer your time. Find out about environmental organizations in your area. Maybe you can help care for injured animals, clean up a habitat, or even answer the phone!	☐	☐
Harness your housepets. If your house pet runs loose it is considered to be an <u>invasive species</u> . In the United States alone, it is estimated that 44 million free-roaming cats kill 4.4 million songbirds every day.	☐	☐
Be a smart shopper. Buy things you really need—not just want—so you don't waste Earth's resources on unnecessary items. Shop for good quality, so you can use toys, clothes, and supplies many times, then pass them on to someone else. When you travel, avoid buying souvenirs that cost an animal its life or might have harmed a habitat.	☐	☐
Reduce, reuse, recycle* ! If you choose not to turn on a light, you save energy. If you reuse a jacket handed down by a brother or sister, you've saved all the resources that would have gone into a new jacket. When you recycle, you save energy, electricity, water, and many other resources, too. Earth-keepers always remember the 3 Rs!	☐	☐
Leave wild things in the wild. Watch wild things, but don't bring them home. Plants and animals often die outside their habitats. Even if they live, they can't reproduce or do their "jobs" in the natural world.	☐	☐
Learn about other cultures. What you do and what you buy can affect other people around the world. Learn about other cultures to discover some new ways to help <u>biodiversity</u> .	☐	☐
Be a role model. Kindly show your friends and family how to take care of the <u>Earth</u> . Soon, they'll want to be like you!	☐	☐



LESSON PLAN: INTEGRATING DRAMA INTO ENVIRONMENTAL EDUCATION

Lesson Development

Class Level: Grades 8–12 / Adult education (adaptable for teacher workshops)

Duration: 40'+40'

Materials: –Students need cameras or their own camera phones for filming short movies (1 camera per group of 3–4 students). The drama exercises require space, either by moving desks and chairs to the sides of the classroom or by using the school gym. Students need comfortable, casual clothing that is appropriate for physical activities.

Preparation

Research on the use of drama as a tool for developing sustainability skills has focused particularly on higher education students, but drama is also highly applicable in primary and secondary education, where it can help students learn more about sustainable development and climate change. Drama is a practice-based learning approach that uses embodied, story-driven, and collaborative activities to develop sustainability awareness, knowledge, and action-oriented mindsets in education (Wall et al., 2025). It often includes methods such as role play, forum theatre (exploring scenarios through pausing and reworking actions), and legislative theatre, where participants work with decision-makers to address real-world issues (Wall et al., 2025).

Despite progress, sustainability education often depends on individual schools or teachers, which limits its wider impact (Education and Training Monitor, 2024). The idea of this lesson is to develop students' sustainability skills through drama-based activities. Drama exercises support students' active participation during the learning process. The purpose is to increase teachers' teaching resources related to education for sustainable development and environmental education, and to help integrate drama-based teaching methods as part of instructional support.



Improvisation exercise

Students are guided into an improvisation exercise. Two students are chosen to come to the front of the class. The teacher has written separate words related to sustainable development and climate change on slips of paper placed in a container. Possible words include, for example, recycle, litter, save electricity, forest fire, melting glacier, cycle, walk, pollute, trash, nature conservation area, bird, ecological, environmentally friendly, wolf, flood, heatwave, bus, drive, second-hand store, recycling center, forest, meadow, river, factory, solar panel, nature trail, organic food, wind turbine, etc.

The class collectively decides what roles the two students will take on in the exercise. Each student, in turn, draws a word from the container, and they must incorporate the word into their improvisation. Once the words have been included in the improvisation, another student from the audience can join the scene, and one of the students who was initially on stage leaves. The audience again decides what role the student will play. The new student draws a new word from the container and incorporates it into the improvisation. This continues until all students have taken part in the improvisation exercise in turn. At the end of the exercise, the class discusses how the improvisation went and whether it was challenging to include concepts related to sustainable development and climate change. If there are only a few students, they can go on stage several times during the activity.



Sustainability Statues

Students are divided into groups of about 5–6. The aim is for the students to freeze into statues and create a tableau as a group based on a given idea. The teacher provides an idea such as climate change, sustainable development, endangered animal, nature conservation, a local farm, a wind turbine, an ecological city, etc. Within about one minute, the students are expected to create a group tableau on the given topic using their imagination. At the end, the students explain the idea behind their tableau. During the activity, the students create approximately six different tableaux.

Telephone game

In this game, students notice how easily a communicated message changes as it is passed from one person to another. The students sit in a circle, and the first student whispers a short message into the ear of the next student. The message can be a fact or statement related to climate change or sustainable development (e.g., “Glaciers are melting and polar bears are in trouble” or “Recycling helps protect the environment”). The message is whispered only once without repetition. Each student passes the message on in turn to the next student. The last student says the message out loud to the class.



Short film

Students are divided into groups of about 3–4. Each group plans and scripts a short film based on a given idea (ideas: Littering is not acceptable, Climate change, A nature conservationist, Eco-hero, Today's eco-friendly choice, Nature is important). The students film their movies using mobile phone cameras or cameras borrowed from the school. Finally, the completed films are watched together as a class, and the students are given the opportunity to tell more about their films.

Sources:

European Commission. (2024). Education and Training Monitor 2024. Publications Office of the European Union. <https://op.europa.eu/>

Wall, T., Österlind, E., Lehtonen, A., et al. (2025). Drama as a transformational capability of sustainability science. Sustainability Science. <https://doi.org/10.1007/s11625-025-01751-2>



LESSON PLAN: SUPPORTING THE DEVELOPMENT OF SUSTAINABILITY COMPETENCIES

Lesson development

Class Level: Grades 8–12 / Adult education (adaptable for teacher workshops)

Duration: 40'+40'

Materials:

- **Mind Map (one per class) A smart board, a chalkboard, or the free Padlet application works well for this activity.**
- **Students should have access to computers or iPads during the lesson.**
- **The instruction posters created during the lesson are printed and displayed on the walls of the classroom or school corridors.**

Preparation: General discussion

The aim of the European Union is to build students' sustainability competencies so that citizens and communities learn to care for the environment and work toward sustainable development. Nevertheless, research shows that only 42% of young people report having had a meaningful opportunity to learn about sustainability in school. (Education and Training Monitor, 2024).

The class discusses climate change. The teacher explains to the students what climate change means. Using a mind map, the students can list concepts and phenomena related to climate change together with the teacher. The teacher explains that every student can, through their everyday choices, contribute to mitigating climate change in daily life. The aim of the lesson is to understand that everyday actions matter in the overall picture and that if every student commits to taking environmental actions, their collective impact can be significant on a wider scale.



The students work in small groups to think about ways individuals can help combat climate change. How can a student personally contribute to fighting climate change? The discussion covers recycling, avoiding food waste, choosing walking, cycling, and public transport, saving energy at home and school, consumption habits, voting, and influencing community actions to mitigate climate change.

Let's discuss together: How can food waste be reduced at home and at school? Why is it beneficial to prefer local and organic food? How do I recycle waste correctly? Why are walking, cycling, and public transport environmentally friendly choices? Is reducing electricity and water consumption an eco-friendly action?

Electricity and water consumption

The teacher asks students to list typical household electrical appliances and writes them on the board. (Light bulb, energy-saving bulb, refrigerator, freezer, oven, computer, television, vacuum cleaner, radio, iPad etc.)

The teacher teaches how much energy different electrical appliances consume. The students work in groups to solve exercises. How much electricity is used if the oven is on for 2 hours? How much energy does a television consume if it is on for 3 hours? What happens if a light bulb is replaced with an energy-efficient one? How much less electricity does the bulb use in a week if it is on, for example, about 4 hours per day? Under the teacher's guidance, the students discuss how they could use less electricity in their everyday choices.



Electricity consumption per hour (approx.)



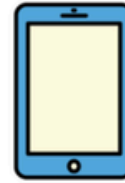
Incandescent light bulb:
~60 W → 0.06 kWh/hour



Energy-saving bulb
(LED/CFL):
~8–12 W →
0.008–0.012 kWh/hour



Television: ~50–150 W →
0.05–0.15 kWh/hour



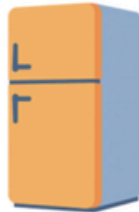
iPad / tablet: ~5–10 W →
0.005–0.01 kWh/hour



Radio: ~5–20 W →
0.005–0.02 kWh/hour



Refrigerator: ~100–200 W
(cycles on/off)
→ ~0.1–0.2 kWh/hour



Freezer: ~150–300 W
(cycles on/off) →
~0.15–0.3 kWh/hour



Oven (electric): ~2000–3000 W →
2–3 kWh/hour



Vacuum cleaner: ~600–2000 W →
0.6–2.0 kWh per hour

Students reflect on their own water consumption. What is water used for? Would it be possible to use water more sparingly? For example, let's consider spending time at a summer cottage without running water. Does washing dishes by hand reduce water consumption? If water is fetched from a well, for example, does it lead to more mindful use of water?



Recycling and wood-waste

Let's review recycling instructions. How do we recycle paper, glass, metal, cardboard, plastic, bio-waste, electronics, medicines, mixed waste, and old clothes? Let's discuss. How can I reduce the amount of waste I generate? How do I minimize food waste during school lunch?

Students can test their recycling skills using the University of Michigan's game on sustainable development and eco-friendly actions. Students can try the "Sort the Trash on Earth" game about correctly recycling waste

<https://seas.umich.edu/admissions/games>

Carbon Footprint

The teacher explains what an individual's personal carbon footprint consists of. The Consumer Footprint Calculator on the European Commission's website offers EU citizens a way to calculate the environmental impacts of their consumption patterns and to evaluate how changes in their lifestyle may affect their personal footprint. The teacher can go through a carbon footprint calculator together with the students, and they can create an example individual, John or Jane Doe, and estimate the size of their carbon footprint together using the calculator.

<https://eplca.jrc.ec.europa.eu/ConsumerFootprint.html>



Games about sustainability

Students may complete Google Earth's "Your Plan, Your Planet" exercise on the impact of everyday choices: <https://www.google.com/earth/education/resources/>

Students can test their recycling skills using the University of Michigan's game on sustainable development and eco-friendly actions. Students can try the "Day in the Life" game about how everyday choices can help reduce their carbon footprint.

<https://seas.umich.edu/admissions/games>

Ideas for green skills in schools

Finally, divide the students into groups of 2–3. Ask each group to create instructions for the school, with images, related to saving energy and water, reducing waste, and recycling. Students can design an instructional poster using the Canva app.

Possible topics include: energy consumption in the classroom, economical use of water, recycling food waste and reducing waste, recycling trash in the classroom, saving paper, and preventing littering in the school environment.



Evaluation

- Each group presents their project in 3 minutes.
- The other groups evaluate the presented project according to the criteria (1-5 points):

Criteria	Point				
	1	2	3	4	5
Are the instructions on the poster understandable and clear?					
Do the instructions support the school's sustainable development?					
Does the poster design serve its purpose?					
Is the design understandable?					
How well do the instructions work in a school context?					

Teacher assessment:

- Group presentations (content, creativity, collaboration)
- Originality and applicability of the design
- Students' correct use of concepts



Sources:

European Commission (2024) Education and Training Monitor 2024 – Comparative report. Luxembourg: Publications Office of the European Union. <https://op.europa.eu/>

European Commission, Joint Research Centre. (n.d.). *Consumer Footprint calculator*. <https://eplca.jrc.ec.europa.eu/ConsumerFootprint.html>

Google. (n.d.). *Google Earth Education resources*. <https://www.google.com/earth/education/resources/>

University of Michigan School for Environment and Sustainability. (n.d.). *Admissions FAQ / Games*. <https://seas.umich.edu/admissions/games>



GREEN GAMES



Eco Empire

Eco Empire is a free online simulation game where players build and manage recycling centers and transform waste into valuable products. It teaches principles of the circular economy, resource management, and sustainable decision-making.

Link: <https://plays.org/eco-empire/>

Learned skills

- Circular Economy Basics: Introduces players to recycling, reusing, and efficient use of resources.
- Problem-Solving Skills: Players develop strategic thinking and decision-making skills to run an effective recycling center.
- Environmental Awareness: Promotes sustainability and encourages eco-friendly practices.



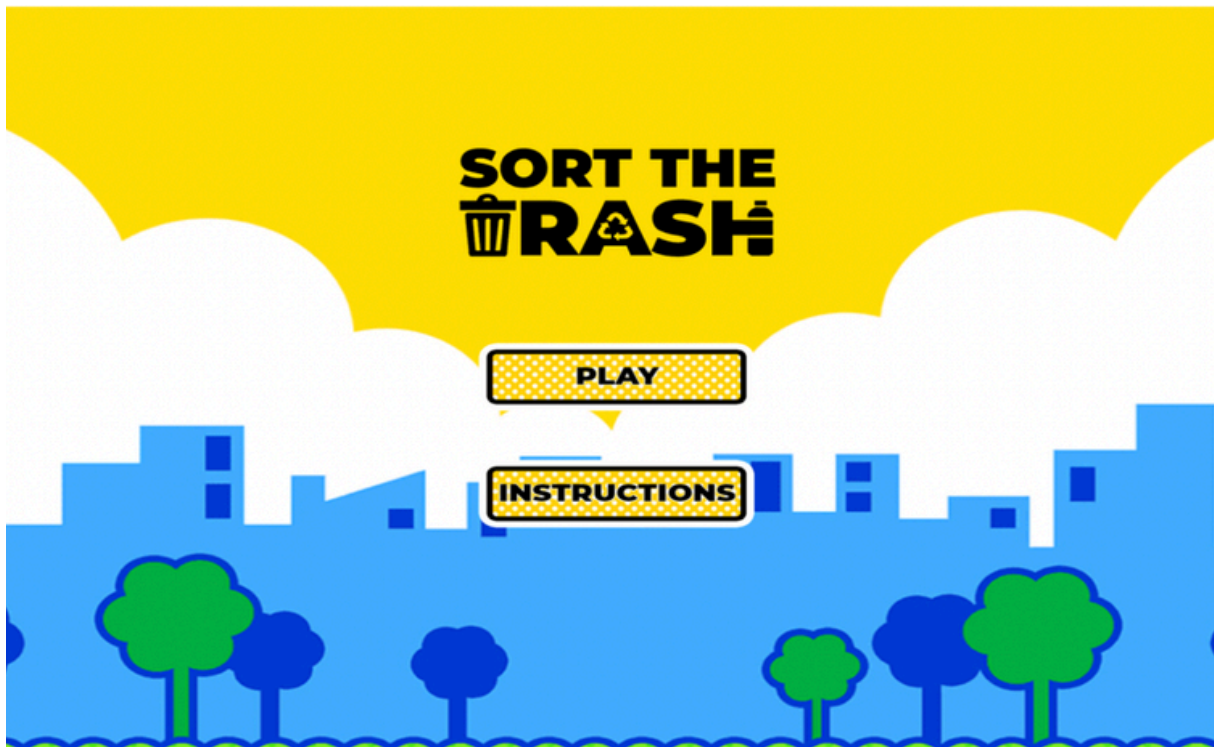
Climate Kids

Climate Kids is a free online platform created by NASA that teaches students about climate change, energy, and ecosystems through interactive games, quizzes, and videos. NASA's Climate Kids provides a comprehensive way for young learners to understand the science of climate change, fostering a generation of informed and environmentally conscious individuals.

Link: <https://www.nasa.gov/stem-content/nasa-climate-kids/>

Learned skills

- Understanding Earth's Systems: Learn about the atmosphere, oceans, weather patterns, and how they interact to shape the planet's climate.
- Promoting Environmental Stewardship: Encourage actions that individuals can take to protect the environment, such as conserving energy and reducing waste.



Sort the trash on Earth

Sort the Trash on Earth is an interactive educational game developed by the University of Michigan's School for Environment and Sustainability. Designed to involve players in understanding recycling practices, the game challenges users to correctly sort various items into correct waste categories.

Link: <https://seas.umich.edu/assets/games/recycle-game/>

Learned skills

- Awareness: Increase understanding of proper waste sorting and recycling practices.
- Engagement: Encourage players to think critically about waste management and its environmental impact.
- Local Relevance: Highlight the importance of adhering to local recycling guidelines and programs.



Day in the Life

Day in the Life is an interactive educational game developed by the University of Michigan's School for Environment and Sustainability. The game immerses players in the daily routines of various individuals, highlighting the environmental impacts of everyday choices. Link:

<https://seas.umich.edu/assets/games/day-in-life/>

Learned skills

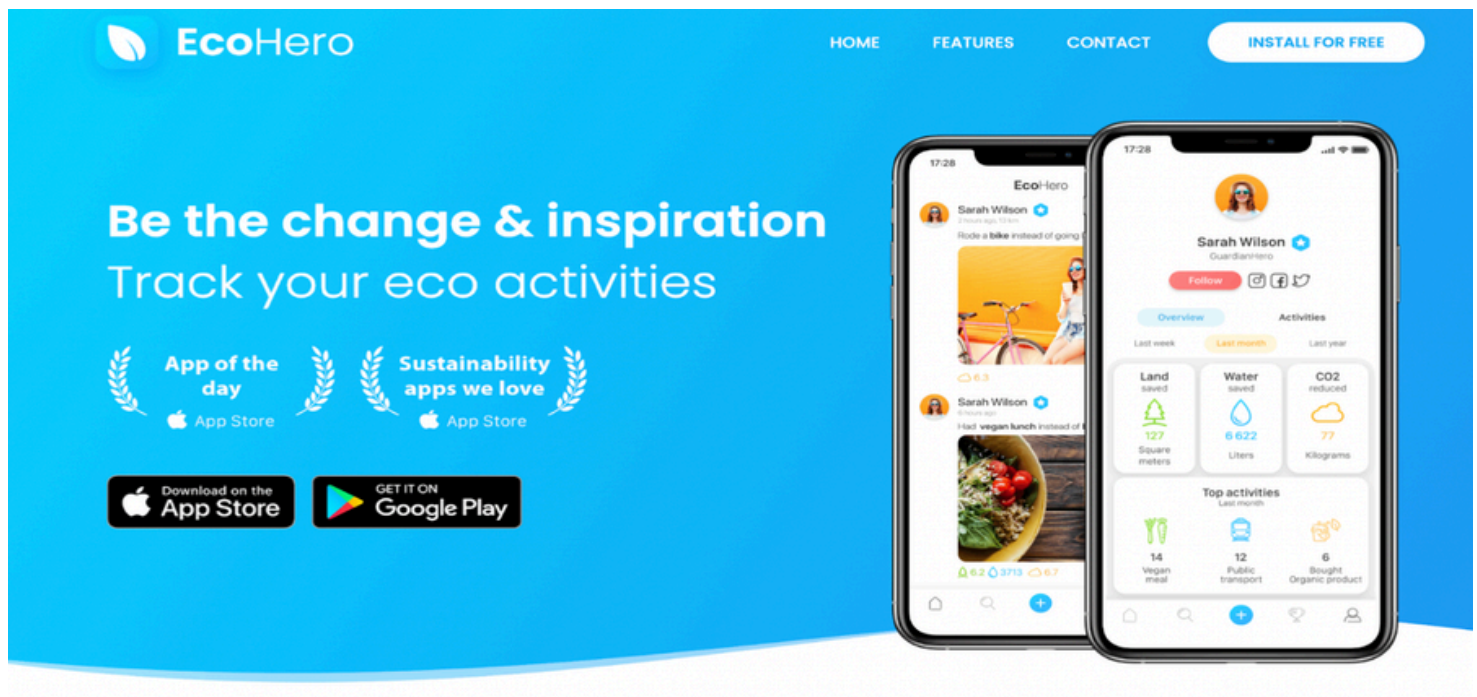
- Awareness: Increase understanding of how daily activities contribute to environmental issues like climate change and resource depletion.
- Encouraging Change: Encourage players to adopt more sustainable practices in their own lives.



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EcoHero

EcoHero is a free digital platform and mobile app that encourages students to take actionable steps to reduce their environmental impact. It allows students to track eco-friendly habits, calculate their carbon footprint, and engage in challenges that promote sustainability. EcoHero uses gamified challenges to motivate students, making learning about sustainability fun. Link: <https://ecohero.app/>

Learned skills

- Promotes hands-on learning and personal responsibility.
- Develops data literacy through tracking and analyzing sustainable actions.
- Reinforces critical thinking about environmental issues and solutions.